MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

SEVENTH SEMESTER BTECH. (E & C) DEGREE REGULAR END SEMESTER EXAMINATION NOV/DEC 2024

SUBJECT: MOTION AND GEOMETRY BASED METHODS IN COMPUTER VISION (ECE4078)

TIME: 3 HOURS **MAX. MARKS: 50**

Instructions to candidates
- Answer **ALL** questions.
- Missing data may be suitably assumed.

Q. No.	Questions	M*	C*	A*	B*
1A.	Describe Speeded Up Robust Features (SURF) detection and descriptor method, with the help of mathematical equations. Compare its advantages and limitations over Scale Invariant Feature Transform (SIFT) method.	5	2	2	4
1B.	For the given object with coordinates at (0,0), (1,2) and (2,1), calculate the new point coordinates obtained after the following 2D transformations: a.uniform scaling by factor 2 b.translation by (3,-1) c.horizontal shearing by factor 0.25	3	1	3	3
1C.	Compute the Local Binary Pattern (LBP) Map of the given 3×3 image, *Img*, using LBP with radius = 1 pixel and neighbours = 8 pixels. Assume zero padding. $Img = \begin{bmatrix} 50 & 100 & 75 \\ 25 & 40 & 100 \\ 100 & 75 & 50 \end{bmatrix}$	2	2	1	3
2A.	Explain how Active Appearance Model (AAM) method is used for object registration in images. Discuss its advantages over Active Shape Model (ASM) method.	5	3	2	4
2B.	Describe the procedure of Iterative Closest Point (ICP) algorithm for image registration, using suitable diagrams.	3	3	1	2
2C.	Discuss how the object scaling and the aperture size of the camera affect the projection in image plane, using a pinhole camera model. Justify your answer with the help of suitable examples.	2	1	2,3	5
3A.	Describe Kalman filter for object tracking in video sequence. With the help of mathematical equations, explain the prediction-correction mechanism for tracking.	5	4	1	2
3B.	Compare and contrast the differential motion analysis of an object in a video sequence using (a) background subtraction, and (b) background estimation using Gaussian mixture models.	3	4	2	4
3C.	State the advantages of using quaternions for range image registration.	2	3	1	2

4A.	Describe the step by step procedure of Dynamic programming for binocular fusion, with the help of suitable diagrams and examples. Discuss its advantages and limitations.	5	5	2	4
4B.	With the help of suitable diagrams, compare the perspective camera model, weak perspective camera model and orthographic projection camera model.	3	5	2	4
4C.	Determine the intersection of a line and plane using homogenous coordinates. Given, $l(t) = (5 - t, 2 + 3t, 1 + 4t, 1)$ and plane $5x + 3y - z = 12$	2	5	1	3
5A.	Discuss in detail the direct and stratified methods for transforming Perspective to Metric reconstruction. Highlight the significance of each method.	5	6	2	2
5B.	Explain Hartley's normalization technique for image 3D reconstruction. Discuss its significance.	3	6	2	2
5C.	Calculate the minimum number of points required for weak perspective Structure from Motion for two cameras. Show the equation used and justify your answer.	2	6	2	3

M*--Marks, C*--CLO, A*--AHEP LO, B* Blooms Taxonomy Level